

April 14, 2026

Editor-in-Chief
Proceedings of the IEEE

Dear Editor,

We submit for your consideration the manuscript:

Self-Adaptive Model-Based Protection for IBR-Penetrated Networks: A Unified Five-Layer Framework

for publication as a *Tutorial and Survey* article in *Proceedings of the IEEE*.

The problem. Inverter-based resources (IBRs) — photovoltaic plants, DFIG wind turbines, battery storage, and virtual synchronous machines — are penetrating distribution and sub-transmission networks at levels that break the fundamental assumptions of classical protection (overcurrent, distance, and differential). IBR current limiters clip fault currents at 1.1–1.2 pu, IBR control laws transition modes during sustained faults, and the aggregate penetration ratio k_{ibr} varies continuously. As a result, classical differential relays with fixed slope settings are losing 7–11 percentage points of zone accuracy when $k_{\text{ibr}} > 0.25$. There is no unified framework that combines online IBR estimation, physics-constrained adaptation, and stability-verified tripping in a single deployable relay algorithm.

Contributions. This paper presents the **Self-Adaptive Model-Based Protection System (SAMBPS)**, a five-layer hierarchical architecture developed over sixteen theoretical contributions and validated in seven independent campaigns:

1. **Unified architecture (C16):** A five-layer framework (signal conditioning, LM estimation, adaptive laws, stability verification, protection decision) with well-defined layer contracts and a single integrating signal — the condition number κ_n from the Levenberg–Marquardt estimator.
2. **κ_n as system integrator:** We prove that the same threshold $\kappa_n = 30$ is simultaneously the gate condition for the adaptive law (C1), the SVD reliability criterion for the LM estimator (C9), the CUEP gate (C13), and the Stage-2 trip veto (L5). This triple-role consistency is proved in an appendix.
3. **Comprehensive theory-to-validation mapping:** All sixteen contributions and seven campaigns are allocated to specific layers via a contribution map (Fig. 2), showing that every layer has at least two independent validation campaigns.
4. **Field demonstration:** 95.5% zone accuracy on 157 real 33 kV DSO fault recordings, compared to 89.2% for classical protection on the same dataset.

Fit to *Proceedings of the IEEE*. *Proceedings of the IEEE* is the primary venue for broad tutorial surveys that unify a research programme and orient the field towards future directions. This paper synthesises five primary research papers (C1, C9, C13, C14, and seven G-series validations) into a coherent framework suitable for readers who want to understand the full SAMBPS system without reading each component paper. The architecture is novel, the field

validation is the most comprehensive to date, and the open problems section identifies concrete next steps for the community.

Suggested reviewers.

- Prof. Arun G. Phadke (Virginia Tech) — wide-area protection, phasor measurement
- Prof. Ali Hooshyar (University of Toronto) — inverter-based resource protection
- Prof. Vijay Vittal (Arizona State University) — transient stability, power system dynamics
- Prof. Petros Ioannou (University of Southern California) — adaptive control, Lyapunov methods

This manuscript is not under review elsewhere. All authors have approved the submission.

Sincerely,

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